

Alert INC-4271 spike in authentication failures and 4xx/5xx errors on the web tier, last hour

8 candidate causes + other; hidden true cause **credential_stuffing**

Both runs: the same Qwen2.5-7B planner, read-only probes, budget (10 cost units, 12 decisions) and independent verifier

The planner chooses every probe

Memory-only: π reads the evidence ledger and picks the next probe itself.

<i>probe</i> $\rightarrow$ <i>observed outcome</i>	<i>cost</i>	<i>used</i>	<i>p(h*)</i>	<i>step</i>
auth_log $\rightarrow$ distributed_failures	■	1/10	████.96	1
auth_log <i>repeat, blocked</i>			████.96	2
source_ips $\rightarrow$ many_residential	■	2/10	████≈1	3
change_ticket $\rightarrow$ none	■	3/10	████≈1	4
auth_log <i>repeat, blocked</i>			████≈1	5
access_pattern $\rightarrow$ normal	■	4/10	████≈1	6
component_versions $\rightarrow$ up_to_date	■	5/10	████≈1	7
threat_intel $\rightarrow$ malicious_campaign	■■■	8/10	████≈1	8
dns_logs $\rightarrow$ normal	■■	10/10	████≈1	9
auth_log <i>refused: budget spent</i>			████≈1	10

No verdict: budget spent, no conclusion

Qwen2.5-7B over all 24 alerts $\times$ 3 repeats: 9/72 verified

p(h)* passed .9 at step 1; 2 steps lost to repeats

The controller chooses, the planner concludes

HESP: probes are picked by argmax EIG/c over legal, unseen probes; π proposes and concludes; a guard and the verifier check the verdict.

<i>probe</i> $\rightarrow$ <i>observed outcome</i>	<i>EIG/c</i>	<i>cost</i>	<i>used</i>	<i>p(h*)</i>
source_ips $\rightarrow$ many_residential <i>E</i>	1.74	■	1/10	████.82
access_pattern $\rightarrow$ normal π : auth_log	0.39	■	2/10	████.94
auth_log $\rightarrow$ distributed_failures <i>E</i>	0.13	■	3/10	████.98
admin_exposure $\rightarrow$ protected	0.06	■	4/10	████.99
component_versions $\rightarrow$ up_to_date	0.03	■	5/10	████.99
change_ticket $\rightarrow$ none	0.02	■	6/10	████≈1
dns_logs $\rightarrow$ normal	0.00	■■	8/10	████≈1
user_activity $\rightarrow$ normal	0.00	■■	10/10	████≈1
finish $\rightarrow$ credential_stuffing with <i>E</i> <i>none</i>				████≈1

✓ Verdict passes the guard and the verifier

Qwen2.5-7B over all 24 alerts $\times$ 3 repeats: 72/72 verified

no legal probe left $\rightarrow$ π concludes and the check passes